

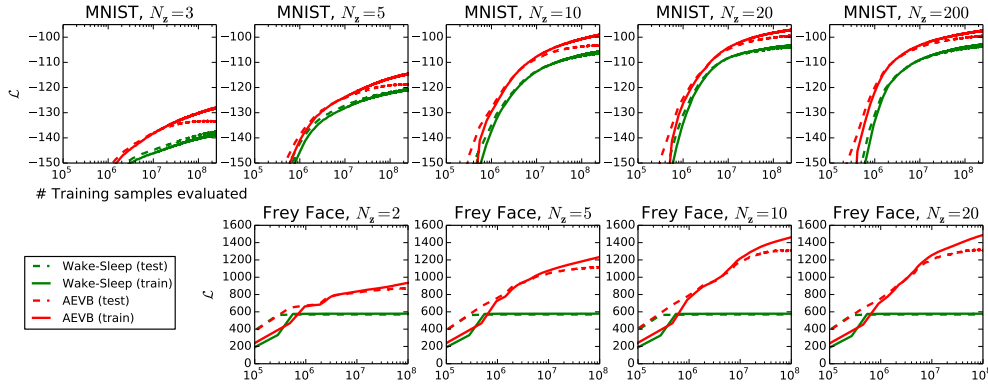


Figure 2: Comparison of our AEVB method to the wake-sleep algorithm, in terms of optimizing the lower bound, for different dimensionality of latent space (N_z). Our method converged considerably faster and reached a better solution in all experiments. Interestingly enough, more latent variables does not result in more overfitting, which is explained by the regularizing effect of the lower bound. Vertical axis: the estimated average variational lower bound per datapoint. The estimator variance was small (< 1) and omitted. Horizontal axis: amount of training points evaluated. Computation took around 20-40 minutes per million training samples with a Intel Xeon CPU running at an effective 40 GFLOPS.

decoder output. Note that with *hidden units* we refer to the hidden layer of the neural networks of the encoder and decoder.

Parameters are updated using stochastic gradient ascent where gradients are computed by differentiating the lower bound estimator $\nabla_{\theta, \phi} \mathcal{L}(\theta, \phi; \mathbf{X})$ (see algorithm 1), plus a small weight decay term corresponding to a prior $p(\theta) = \mathcal{N}(0, \mathbf{I})$. Optimization of this objective is equivalent to approximate MAP estimation, where the likelihood gradient is approximated by the gradient of the lower bound.

We compared performance of AEVB to the wake-sleep algorithm [HDFN95]. We employed the same encoder (also called recognition model) for the wake-sleep algorithm and the variational auto-encoder. All parameters, both variational and generative, were initialized by random sampling from $\mathcal{N}(0, 0.01)$, and were jointly stochastically optimized using the MAP criterion. Stepsizes were adapted with Adagrad [DHS10]; the Adagrad global stepsize parameters were chosen from $\{0.01, 0.02, 0.1\}$ based on performance on the training set in the first few iterations. Minibatches of size $M = 100$ were used, with $L = 1$ samples per datapoint.

Likelihood lower bound We trained generative models (decoders) and corresponding encoders (a.k.a. recognition models) having 500 hidden units in case of MNIST, and 200 hidden units in case of the Frey Face dataset (to prevent overfitting, since it is a considerably smaller dataset). The chosen number of hidden units is based on prior literature on auto-encoders, and the relative performance of different algorithms was not very sensitive to these choices. Figure 2 shows the results when comparing the lower bounds. Interestingly, superfluous latent variables did not result in overfitting, which is explained by the regularizing nature of the variational bound.

Marginal likelihood For very low-dimensional latent space it is possible to estimate the marginal likelihood of the learned generative models using an MCMC estimator. More information about the marginal likelihood estimator is available in the appendix. For the encoder and decoder we again used neural networks, this time with 100 hidden units, and 3 latent variables; for higher dimensional latent space the estimates became unreliable. Again, the MNIST dataset was used. The AEVB and Wake-Sleep methods were compared to Monte Carlo EM (MCEM) with a Hybrid Monte Carlo (HMC) [DKPR87] sampler; details are in the appendix. We compared the convergence speed for the three algorithms, for a small and large training set size. Results are in figure 3.

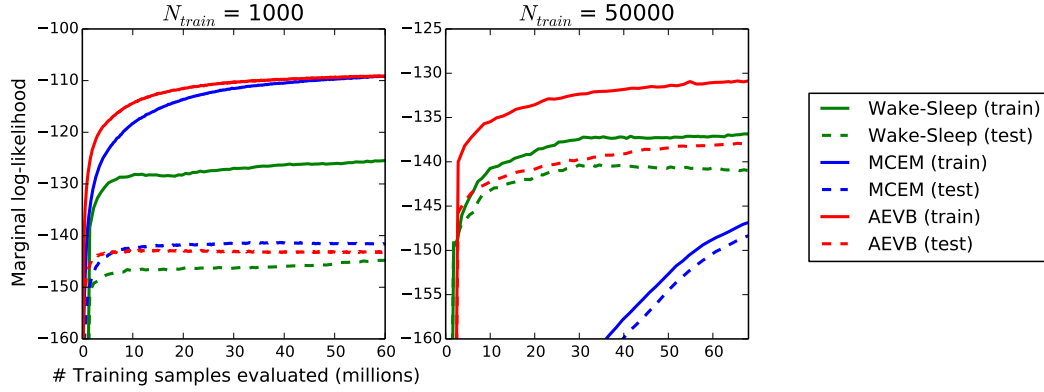


Figure 3: Comparison of AEVB to the wake-sleep algorithm and Monte Carlo EM, in terms of the estimated marginal likelihood, for a different number of training points. Monte Carlo EM is not an on-line algorithm, and (unlike AEVB and the wake-sleep method) can't be applied efficiently for the full MNIST dataset.

Visualisation of high-dimensional data If we choose a low-dimensional latent space (e.g. 2D), we can use the learned encoders (recognition model) to project high-dimensional data to a low-dimensional manifold. See appendix A for visualisations of the 2D latent manifolds for the MNIST and Frey Face datasets.

6 Conclusion

We have introduced a novel estimator of the variational lower bound, Stochastic Gradient VB (SGVB), for efficient approximate inference with continuous latent variables. The proposed estimator can be straightforwardly differentiated and optimized using standard stochastic gradient methods. For the case of i.i.d. datasets and continuous latent variables per datapoint we introduce an efficient algorithm for efficient inference and learning, Auto-Encoding VB (AEVB), that learns an approximate inference model using the SGVB estimator. The theoretical advantages are reflected in experimental results.

7 Future work

Since the SGVB estimator and the AEVB algorithm can be applied to almost any inference and learning problem with continuous latent variables, there are plenty of future directions: (i) learning hierarchical generative architectures with deep neural networks (e.g. convolutional networks) used for the encoders and decoders, trained jointly with AEVB; (ii) time-series models (i.e. dynamic Bayesian networks); (iii) application of SGVB to the global parameters; (iv) supervised models with latent variables, useful for learning complicated noise distributions.

References

- [BCV13] Yoshua Bengio, Aaron Courville, and Pascal Vincent. Representation learning: A review and new perspectives. 2013.
- [BJP12] David M Blei, Michael I Jordan, and John W Paisley. Variational Bayesian inference with Stochastic Search. In *Proceedings of the 29th International Conference on Machine Learning (ICML-12)*, pages 1367–1374, 2012.
- [BTL13] Yoshua Bengio and Éric Thibodeau-Laufer. Deep generative stochastic networks trainable by backprop. *arXiv preprint arXiv:1306.1091*, 2013.
- [Dev86] Luc Devroye. Sample-based non-uniform random variate generation. In *Proceedings of the 18th conference on Winter simulation*, pages 260–265. ACM, 1986.
- [DHS10] John Duchi, Elad Hazan, and Yoram Singer. Adaptive subgradient methods for online learning and stochastic optimization. *Journal of Machine Learning Research*, 12:2121–2159, 2010.
- [DKPR87] Simon Duane, Anthony D Kennedy, Brian J Pendleton, and Duncan Roweth. Hybrid monte carlo. *Physics letters B*, 195(2):216–222, 1987.
- [GMW13] Karol Gregor, Andriy Mnih, and Daan Wierstra. Deep autoregressive networks. *arXiv preprint arXiv:1310.8499*, 2013.
- [HBWP13] Matthew D Hoffman, David M Blei, Chong Wang, and John Paisley. Stochastic variational inference. *The Journal of Machine Learning Research*, 14(1):1303–1347, 2013.
- [HDFN95] Geoffrey E Hinton, Peter Dayan, Brendan J Frey, and Radford M Neal. The “wake-sleep” algorithm for unsupervised neural networks. *SCIENCE*, pages 1158–1158, 1995.
- [KRL08] Koray Kavukcuoglu, Marc’Aurelio Ranzato, and Yann LeCun. Fast inference in sparse coding algorithms with applications to object recognition. Technical Report CBLL-TR-2008-12-01, Computational and Biological Learning Lab, Courant Institute, NYU, 2008.
- [Lin89] Ralph Linsker. *An application of the principle of maximum information preservation to linear systems*. Morgan Kaufmann Publishers Inc., 1989.
- [RGB13] Rajesh Ranganath, Sean Gerrish, and David M Blei. Black Box Variational Inference. *arXiv preprint arXiv:1401.0118*, 2013.
- [RMW14] Danilo Jimenez Rezende, Shakir Mohamed, and Daan Wierstra. Stochastic back-propagation and variational inference in deep latent gaussian models. *arXiv preprint arXiv:1401.4082*, 2014.
- [Row98] Sam Roweis. EM algorithms for PCA and SPCA. *Advances in neural information processing systems*, pages 626–632, 1998.
- [SK13] Tim Salimans and David A Knowles. Fixed-form variational posterior approximation through stochastic linear regression. *Bayesian Analysis*, 8(4), 2013.
- [SL10] Ruslan Salakhutdinov and Hugo Larochelle. Efficient learning of deep boltzmann machines. In *International Conference on Artificial Intelligence and Statistics*, pages 693–700, 2010.
- [VLL⁺10] Pascal Vincent, Hugo Larochelle, Isabelle Lajoie, Yoshua Bengio, and Pierre-Antoine Manzagol. Stacked denoising autoencoders: Learning useful representations in a deep network with a local denoising criterion. *The Journal of Machine Learning Research*, 9999:3371–3408, 2010.

A Visualisations

See figures 4 and 5 for visualisations of latent space and corresponding observed space of models learned with SGVB.